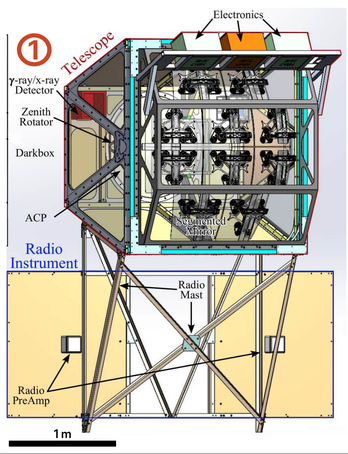
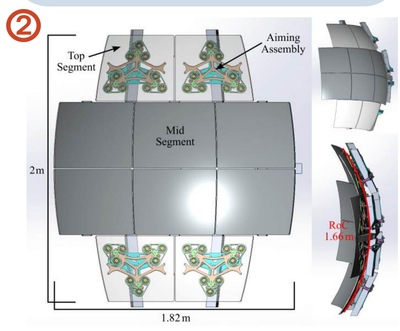
Flexure Mechanical Testing for PBR

Poster By Claire Delavan for the JEM-EUSO Collaboration

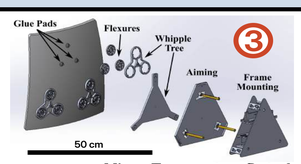
Assisted by Brian Gockel and Eric Mayotte

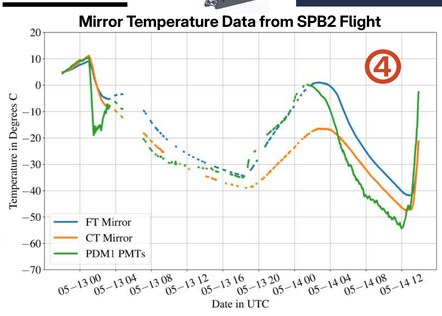


Background



1. The Probe Of Extreme Multi-Messenger Astrophysics (POEMMA) Balloon with Radio (PBR) will study cosmic rays, neutrinos, and high-altitude horizontal air-showers from 33 km above Earth. [1]
2. PBR uses 10 mirrors to direct the particles it collects into the two main detectors. [1]
3. The mirrors are held by Kovar glue pads that then attach to flexures, allowing for thermal contraction. [2]
4. Temperature data from a previous mission shows an expected ΔT of ~65 °C from 10 °C to -55 °C. [3]



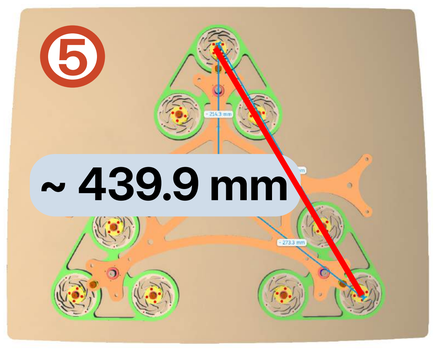


5. The flexure must absorb the thermal contraction with a safety factor of 1.4. [4]

$$D = \Delta T \times (\alpha_1 - \alpha_2) \times L$$

Thermal contraction:
100 x (23.6E-6 - 3.3E-6) x 439.9 ≈ 0.9 mm

Apply saftey factor:
~0.9 x 1.4 ≈ 1.26 mm

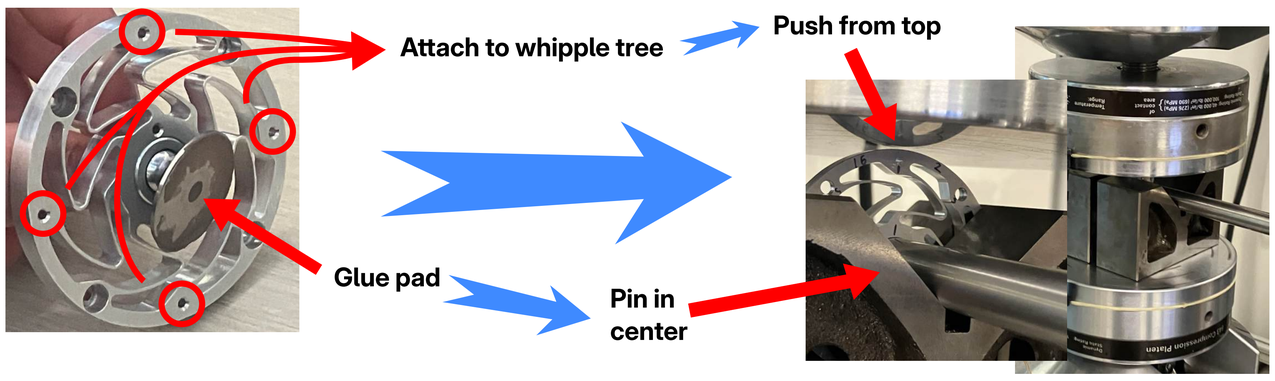


Big Goal: Verify the flexure design to approve for manufacturing, then batch test manufactured flexures.

Geometry Characterization

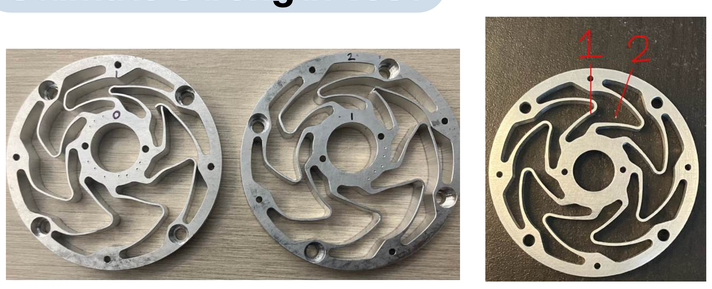
Goal: See failure modes & find spring constant

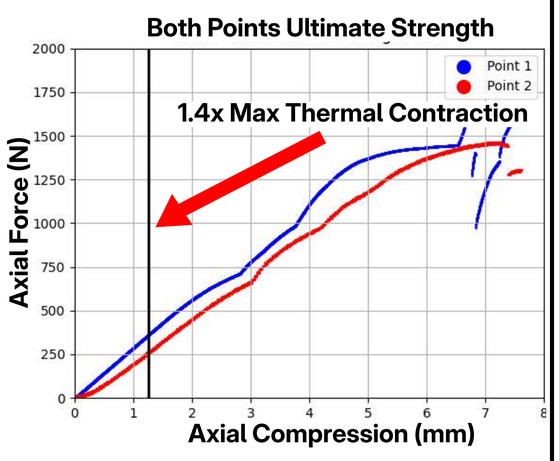
Test Setup



Glue pad in the center and flexure secured to the whipple tree at the edge → Suspend the flexure between V-blocks with a steel rod through the center and press from the top

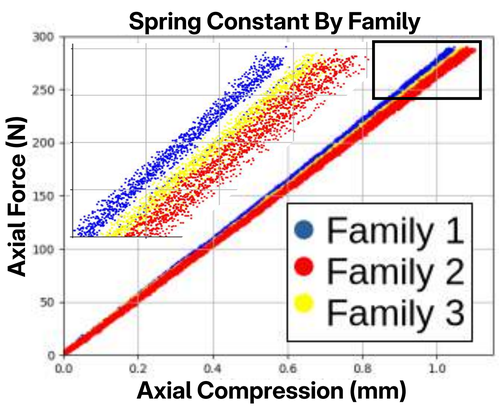
Ultimate Strength Test





The flexures move the required maximum distance at room temperature

Spring Constant Test



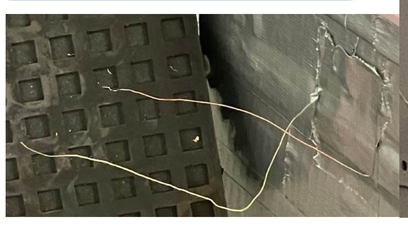
$$k = F / D$$

Family	Average spring constant (N/mm)	Standard Deviation (N/mm)
1	274.53	2.47
2	264.88	2.79
3	268.09	2.14

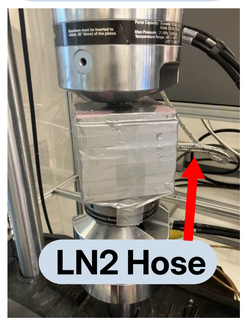
- Pressed on all 16 definable points on 3 flexures
- Each point type is a distinct family by Student's T-Test

Low Temperature Characterization

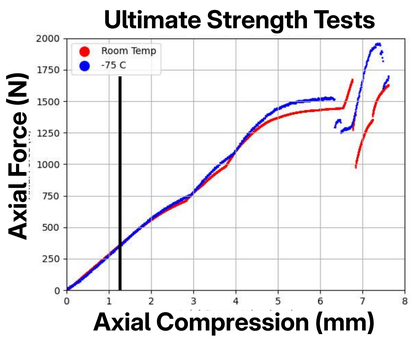
Goal: Observe behavior at flight conditions

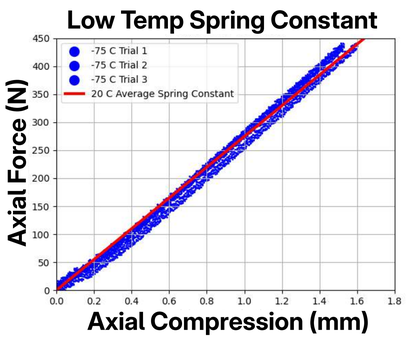


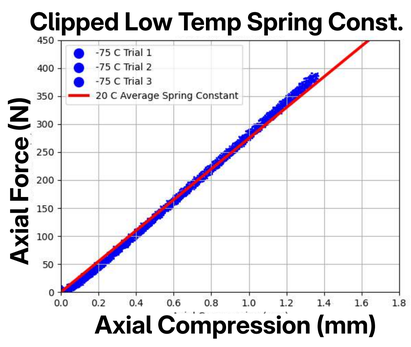
Test Setup



Point 1 Ultimate Strength & Spring Constant Tests





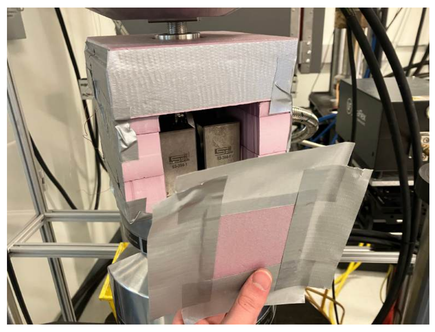


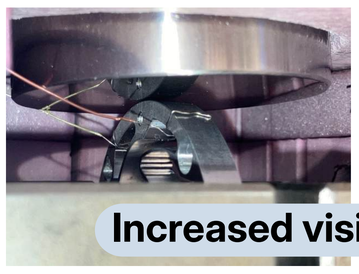
- Plastic deformation after 1.26 mm
- Validated for production

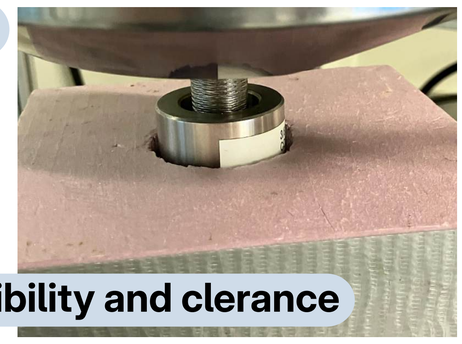
Unexpected non-linearity

Spring constant of 279 N/mm

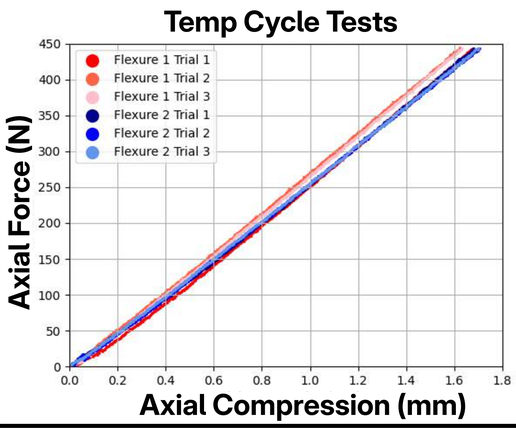
Final Test Setup







Increased visibility and clerance



Temperature Cycling Test

Less non-linearity

No drift between trials

Data is consistent between flexures

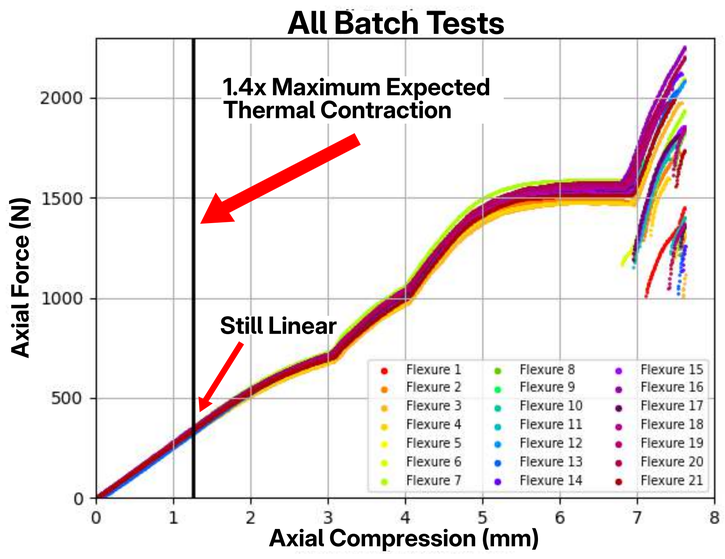
Same spring constant as clipped data

Conclusions

Measured the ultimate strength of every tenth production flexure

All elastically deforming past 1.26 mm

No drift in data from tool wear in manufacturing



Glue bond strength:

k ≈ 279 N/mm @ -75 °C
D ≈ 0.9 mm

k x D ≈ 252 N

Weakest glue bond broke at 1775 N [3]

Passes NASA required 2.0 safety factor [4]

References

[1] JEM-EUSO Collaboration (Adams et al.), "POEMMA-Balloon with Radio: A multi-messenger, multi-detector balloon payload," arXiv:2601.19997 [astro-ph.IM], submitted to JCAP (2026).

[2] E. Mayotte et al., [JEM-EUSO Coll.], The Optical and Mechanical Design of POEMMA Balloon with Radio, PoS(ICRC2025).

[3] L. Wanner et al., [JEM-EUSO Coll.], *Mirror Assembly Glue Bond Testand Verification Plan for POEMMA-Balloon with Radio.*

[4] Columbia Scientific Balloon Facility. Structural Requirements and Recommendations for Balloon Gondola Design (820-PG-8700.0.1A). CSBF Directives Technical Report 820-PG-8700.0.1A,NASA Columbia Scientific Balloon Facility, Palestine, TX, November 2024. Effective date: November 5, 2024. Expires November 5, 2029.

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